A4: Finite-Rod Model and Validity Checks

Part 1. Apply Fourier's law

For steady one-dimensional conduction through a uniform layer, Fourier's law is

$$q'' = -k\frac{dT}{dx}.$$

For constant k and a prescribed heat flux, integration across thickness Δx gives the temperature-drop magnitude

$$\Delta T = \frac{q''\Delta x}{k}.$$

The supplied Problem 1.3 work uses $q''\Delta x = 50\ \mathrm{W\,m^{-1}}$. Thus $\Delta T = (50\ \mathrm{W\,m^{-1}})/k$. The units check is

$$\frac{\mathrm{W\,m^{-1}}}{\mathrm{W\,m^{-1}\,K^{-1}}} = \mathrm{K}.$$

Using the conductivities recorded on the supplied calculation sheet:

Material	$k\ \left[\mathrm{W\,m^{-1}\,K^{-1}}\right]$	ΔT **[K]**
Silver	427	0.117
Aluminum	237	0.211
Steel	54	0.926
Ice	1.88	26.6
Spruce	0.11	455
Insulation	0.071	704
Aerogel	0.024	2.08×10^3

The very large drops for spruce, insulation, and aerogel are physically unreasonable as literal predictions: such extremes would invalidate the constant-property, intact-material assumptions (and could damage the material). The calculation sheet also notes that ice starting at -40°C would reach only about -13.4°C, still below its melting point.

Part 2. Build the stationary finite-rod model

Let T_∞ be the uniform ambient temperature and define the excess temperature $\theta(x) = T(x) - T_\infty$. Consider a differential rod segment of length dx, cross-sectional area A, and perimeter P. At stationarity, energy conservation requires axial heat entering minus heat leaving to equal the heat lost from the side:

$$\dot{Q}_x - \dot{Q}_{x+dx} - hP\,dx\,[T(x) - T_\infty] = 0.$$

The side-loss term has units $(\mathrm{W\,m^{-2}\,K^{-1}})(\mathrm{m})(\mathrm{K}) = \mathrm{W\,m^{-1}}$; including dx gives watts. Dividing by dx and taking $dx \to 0$,

$$\frac{d\dot{Q}}{dx} + hP(T - T_\infty) = 0.$$

Fourier's law for axial conduction is

$$\dot{Q} = -kA\frac{dT}{dx},$$

with units $(\mathrm{W\,m^{-1}\,K^{-1}})(\mathrm{m^2})(\mathrm{K\,m^{-1}}) = \mathrm{W}$. For constant k and A, substitution gives

$$\frac{d^2\theta}{dx^2} - m^2\theta = 0, \qquad m^2 = \frac{hP}{kA}.$$

Here $hP/(kA)$ has units $\mathrm{m^{-2}}$, so m has units $\mathrm{m^{-1}}$. For a circular rod, $P/A = 4/d$, hence

$$m = \sqrt{\frac{4h}{kd}}.$$

The differential equation has the general exponential solution $\theta = C_1 e^{mx} + C_2 e^{-mx}$. It is convenient to write it relative to the tip as

$$\theta(x) = C\cosh[m(L - x)] + D\sinh[m(L - x)].$$

The two boundary conditions for the finite, insulated-tip model are:

1. **Prescribed base temperature:** $T(0) = T_b$, or $\theta(0) = \theta_b = T_b - T_\infty$. The first rod thermistor hole is the analysis base; its measured temperature sets the model's base value.
2. **Insulated tip:** $dT/dx|_{x\,=\,L} = 0$, equivalently $\dot{Q}(L) = -kA\,dT/dx|_L = 0$. No heat leaves through the end face.

The tip condition sets $D = 0$, and the base condition sets $C = \theta_b/\cosh(mL)$. Therefore

$$\frac{\theta(x)}{\theta_b} = \frac{\cosh[m(L - x)]}{\cosh(mL)}.$$

Part 3. Make a provisional finite-length prediction

The course rod is 6061-T6 aluminum. I use the ambient-property bracket $155 \le k \le 167\ \mathrm{W\,m^{-1}\,K^{-1}}$: evaluating the NIST 6061-T6 conductivity fit over 293–300 K gives approximately 154.4–155.3 $\mathrm{W\,m^{-1}\,K^{-1}}$, while the Module 7 README uses $167\ \mathrm{W\,m^{-1}\,K^{-1}}$ as its planning value. This bracket is a prediction input, not a measured conductivity. The NIST fit's data range ends at 300 K, so the bracket should not be read as a precision characterization at elevated rod temperatures.

For $h = 10\ \mathrm{W\,m^{-2}\,K^{-1}}$, $d = 0.011$ m, and $L = 0.283$ m, the circular-rod expression gives

$$\begin{aligned} m &= \sqrt{\tfrac{4h}{kd}} = 4.67\text{–}4.84\ \mathrm{m^{-1}}, \\ \tfrac{1}{m} &= 0.206\text{–}0.214\ \mathrm{m}, \\ mL &= 1.321\text{–}1.371 \quad \text{(dimensionless)}. \end{aligned}$$

The values $d = 0.011$ m, $L = 0.283$ m, and the planned sensor positions $x = (0,\ 0.025,\ 0.050,\ 0.100,\ 0.150)$ m are from the Module 7 README; the position uncertainty is approximately ±0.003 m. The semi-infinite solution follows from the finite solution:

$$\frac{\theta_{\text{fin}}(x)}{\theta_b} = e^{-mx}\,\frac{1+e^{-2m(L-x)}}{1+e^{-2mL}} \underset{mL\to\infty}{\overset{x\ \text{fixed}}{\longrightarrow}} e^{-mx}.$$

Thus Problem 4.12's semi-infinite limit is $\theta_\infty(x) = \theta_b e^{-mx}$. To assess its error, define the relative error in excess temperature against the finite insulated-tip prediction:

$$\epsilon_T(x) \equiv \frac{|\theta_\infty(x) - \theta_{\text{fin}}(x)|}{\theta_{\text{fin}}(x)}.$$

The following values use the two endpoints of the k bracket; entries are rounded to 0.01 percentage point.

Planned x [m]	ϵ_T **at** $k = 155$ **[%]**	ϵ_T **at** $k = 167$ **[%]**
0.000	0.00	0.00
0.025	1.63	1.72
0.050	3.64	3.81
0.100	9.01	9.31
0.150	16.56	16.89

For a separate check of the tip-boundary approximation in Lienhard Example 4.8 and Problem 4.20, take the tip coefficient to equal the provisional side coefficient, $h_L = h = 10\ \text{W m}^{-2}\,\text{K}^{-1}$. The exact convecting-tip solution has

$$\frac{\theta_{\text{conv}}(x)}{\theta_b} = \frac{\cosh[m(L-x)] + \beta\sinh[m(L-x)]}{\cosh(mL) + \beta\sinh(mL)}, \qquad \beta = \frac{h_L}{mk} = 0.0128\text{–}0.0133.$$

Comparing this to the insulated-tip solution at the measured positions, the largest profile difference is only about 0.41% at $x = 0.150$ m (about 0.22% at $x = 0.100$ m). The usual corrected-length approximation gives

$$L_c = L + \frac{A}{P} = L + \frac{d}{4} = 0.28575\ \text{m};$$

its profile differs from the exact convecting-tip solution by less than 0.001% at these sensor positions. The insulated-tip assumption is therefore a small correction compared with the semi-infinite approximation error for this rod.

Using a 5% maximum relative error as the adequacy criterion, the semi-infinite model is adequate at the first three planned positions, but not over the full measured region: the errors at 0.100 m and 0.150 m exceed 5%. Even with a looser 10% criterion, the farthest sensor fails. This prediction depends on the listed positions, $L = 0.283$ m, $d = 0.011$ m, $h = 10\ \text{W m}^{-2}\,\text{K}^{-1}$, the adopted k bracket, and the stated error definition; it is not a measurement-based validity check.

Part 4. Make a provisional one-dimensional prediction

The transverse Biot number compares radial conduction resistance with side convection resistance. For a circular cross section,

$$\text{Bi}_\perp = \frac{h(A/P)}{k} = \frac{hd}{4k}.$$

Its units cancel: $(\text{W m}^{-2}\,\text{K}^{-1})(\text{m})/(\text{W m}^{-1}\,\text{K}^{-1}) = 1$. With the same h, d, and k bracket,

$$\mathrm{Bi}_\perp = 1.65 \times 10^{-4}\text{–}1.77 \times 10^{-4} \ll 0.1.$$

Using the usual $\mathrm{Bi}_\perp < 0.1$ lumped-cross-section criterion, each cross section is expected to be represented adequately by one temperature. This radial conclusion is independent of the finite-length conclusion in Part 3.

Application — electronic heat sink. Forced airflow gives the larger convective coefficient h than still air. Since $m = \sqrt{hP/(kA)}$, increasing h increases m and decreases the decay length $1/m = \sqrt{kA/(hP)}$: the temperature excess decays over a shorter distance along the heat sink.

References and inputs

- Lienhard and Lienhard, *A Heat Transfer Textbook*, Section 4.5 and Problems 1.3, 4.12, and 4.20 (including Example 4.8).
- Module 7 README, for rod geometry, sensor positions, position uncertainty, and the $k = 167\ \mathrm{W\,m^{-1}\,K^{-1}}$ planning value.
- NIST 6061-T6 aluminum thermal-conductivity data, evaluated over its reported 293–300 K interval.
- Problem 1.3 conductivity values and intermediate calculations transcribed from the supplied `LAB.pdf`.